LESSON 3.7

Ratio Reasoning: Convert Measurements Between Systems

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use ratio reasoning to convert measurements between the customary and metric systems.

WHY THIS MATTERS The academy's kitchen orders from suppliers in two different countries. One ships in pounds and quarts, the other in kilograms and liters. Compare measurements by converting with the appropriate factor. Some factors are exact; rounded factors give approximate results. One supplier ships 10 pounds of rice and the other ships 5 kilograms. Use 1 pound ≈ 0.45 kilogram to find which shipment is more.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Conversion factor *Spanish: Factor de conversión*	The ratio that links a unit in one system to a unit in the other, such as 1 mile to about 1.609 kilometers.	1 m ≈ 3.28 ft
Customary system *Spanish: Sistema usual*	The measurement system used in the United States: inches, feet, miles, ounces, pounds, cups, quarts, gallons.	in · ft · mi · oz · lb · qt · gal
Metric system *Spanish: Sistema métrico*	The measurement system used in most of the world, built on tens: centimeters, meters, kilometers, grams, kilograms, liters.	cm · m · km · g · kg · mL · L
Approximately *Spanish: Aproximadamente*	Close to, but not exactly equal. A rounded conversion factor gives an approximate result, written with ≈; an exact conversion uses =.	≈ means “about”
Ratio table *Spanish: Tabla de razones*	A table of equal ratios used to scale a conversion factor up to the amount you actually need.	km: 1, 10, 100 · mi: 0.6, 6, 60
Reasonableness *Spanish: Sentido lógico*	Checking that the size of an answer makes sense for the situation.	bigger unit → smaller number

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A bag weighs 44 pounds. About how many kilograms? (1 kg ≈ 2.2 lb)

1. A kilogram is heavier, so the number gets smaller: divide.
2. 44 ÷ 2.2 = 20.
3. Check: 20 × 2.2 = 44.

Answer: 44 lb ≈ 20 kg

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Terrance needs flea medicine for a dog weighing 31 pounds, but the packages are labeled in kilograms.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I know 1 _____ ≈ _____ _____ .
- A _____ is bigger than a _____ , so I _____ .
- My answer is about _____ _____ .
- This is reasonable because _____ .

Word bank convert | conversion factor | approximately | about | reasonable | bigger | smaller | multiply | divide | meter | mile | kilogram

WATCH OUT!

A common mistake

Compare measurements only after converting to a common unit. Also check whether the conversion factor is exact or rounded: 1 inch = 2.54 cm and 1 international mile = 1.609344 km are exact

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

1 kilometer ≈ 0.6 mile. About how many miles is 5 kilometers?

(A) 3 miles
(B) 5.6 miles
(C) 8 miles
(D) 30 miles

STEPS TO TRY
1. Write the conversion factor as a ratio: 1 : 0.6.
2. Multiply both parts by 5.
3. 0.6 × 5 = ?

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

1 pound ≈ 0.45 kilogram. About how many kilograms is 10 pounds?

(A) 4.5 kg

(B) 10.45 kg

(C) 22 kg

(D) 45 kg

STEPS TO TRY

1. The ratio is 1 lb : 0.45 kg.
2. Multiply both parts by 10.
3. 0.45 × 10 = ?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which pair of units comes from two DIFFERENT measurement systems?

(A) Inches and feet

(B) Cups and quarts

(C) Grams and kilograms

(D) Miles and kilometers

STEPS TO TRY

1. Sort each unit as customary or metric.
2. Metric units include meter, gram, liter and their prefixes.
3. Which pair has one of each?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A sign in Mexico gives a distance of 80 kilometers. Using 1 km ≈ 0.6 mi, about how far is that in miles?

(A) About 8 miles

(B) About 48 miles

(C) About 80 miles

(D) About 133 miles

STEPS TO TRY

1. Scale the ratio 1 : 0.6 up to 80 kilometers.
2. Multiply 0.6 by 80.
3. Should the miles number be bigger or smaller than 80?

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Why does 1 inch = 2.54 cm use =, while 1 mile ≈ 1.609 km uses ≈? What does the rounding change?

SENTENCE STARTER The inch factor is ___ by definition. The mile factor shown is ___, so I write ___ for its result.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help